A map is a phase without a clock: what path-integrated attention learns, and what decides whether training finds it

Main paper, draft 2 (2026-10-21), from `docs/paper/expansion/SYNTHESIS_DRAFT2_PLAN.md`. *Appendix:* `docs/paper/MAIN_APPENDIX2.md`. *Word counts, the source of every number, deviations from the plan and open items:* `docs/paper/MAIN_DRAFT2_notes.md`. *Delete this line before submission.*

Abstract

Path-integrated rotary attention replaces RoPE's token index with a learned phase that sums a content-dependent step per token. We give an account of what such models learn and of what decides whether training finds a map. On 1024-move torus walks, where a map-free rule that retraces the last run of moves scores 0.843, one path-integrated layer scores 0.998 (8 of 8 seeds solved) against 0.731 for index RoPE. The rotation carries the prefix sum of the moves, so one layer compares positions at any gap; index attention must build that sum in its residual stream, and at matched depth one path layer is worth about three index layers on Dyck-2 and on a biased 1D walk, and more than four on the torus. We show that a map code is a phase without a clock, the phase an out-and-back adds, and bound how much a clock erodes the position match one attention pass can make as the moves since the last visit grow. Two choices decide whether training removes the clock. A signed step at per-head rank above the map's dimension has two routes that remove the clock and leave the map intact; at rank equal to the dimension no channel can be clock-free and a full map while a clock remains; a non-negative step has neither route from a small start. In line with this, rank 2 solves the torus on at most 2 of 8 seeds against 8 of 8 at rank 4, and non-negative steps on 0 of 8 against 8 of 8 for signed steps, although both constrained classes represent the map exactly and keep or re-find an installed solution. On a ring the sign gap holds at each step's best learning rate and vanishes when only the readout, not the increment, is non-negative. Told in English, the walk is learned by one path layer (0.979 vs 0.505), which cancels opposite directions.

1. Introduction

Rotary position encodings are integrators. Index RoPE (Su et al. 2024) adds a constant angle per token; content-dependent variants let each token set its own step (CARoPE, Veisi et al. 2025; CoPE, Golovneva et al. 2024; Selective RoPE, Movahedi et al. 2026; Mamba-3, Lahoti et al. 2026; MapFormer, Rambaud et al. 2026). When the step is driven by actions, the summed phase is a position on a map, and attention compares net displacements instead of token distances. MapFormer built this into a transformer and showed that it helps navigation, generalises to sparser and denser walks, extrapolates on Dyck-2 with one layer, and learns opposite latents for opposite actions. We take that as the starting point. What we add is a measurement of how much of the task needs a map, the exchange rate between path integration and index depth together with its mechanism, a theory of what decides whether training finds the map, and the same mechanism at work on navigation told in words.

The account turns on one decomposition. Split the phase a channel gains over an interval into a part that depends on net displacement, the *map*, and a part that grows with the number of moves whatever their direction, the *clock*. A map code is one whose used channels carry no clock: an out-and-back returns the phase to where it started. Path integration writes the prefix sum of the moves into the rotation, so a clock-free code lets one attention layer find the same place at any gap and across the wrap of the torus. A clock works against this: its phase error grows with the number of moves since the last visit, and Proposition 2 bounds the position match it leaves to one attention pass while that error stays below pi. Every model we see fail, index attention, a rank-2 content-to-angle map and a non-negative step, shows the matching measured signature: it keeps short-gap revisits and loses long-gap and wrap-only ones (Fig. 2).

Depth does not buy the map. Four index layers, or one block looped four times, learn local return rules that reach a few direction runs, and never find the global count that would give them the map (Sec. 6). So the map has to come from the step map, and whether training finds it depends on whether training can remove the clock. A signed step can remove it collectively: driving the latents of opposite moves to sum to zero cancels every channel's clock at once and leaves the map untouched. Above rank D, the map's dimension, each channel can also stop reading the clock on its own, at no cost to the map. A non-negative step has neither route from a small start, and at rank D no channel can be both clock-free and a full map while a clock remains (Proposition 4). The measurements follow this inventory on both axes (Fig. 1): the constrained classes represent the map exactly, keep or re-find an installed solution, and from scratch rarely find it, and the runs that fail from a small start end with a clock of the order of their map or, for a few rank-2 runs, with a map that has lost a dimension.

Contributions.

1. *Measuring map use.* Map-free floors (blank, n-gram, retrace) and revisit strata (short-gap, long-gap, wrap-only) are defined once and used throughout. A retrace rule scores 0.934 on MapFormer's 128-move walks and 0.843 at 1024 moves, where one path layer scores 0.998 against index RoPE's 0.731 (8/8 vs 0/8 SOLVED, p 0.0002) and only the path model holds long-gap and wrap-only revisits.
2. *Why one path layer suffices, and what depth buys index attention.* The prefix sum is in the rotation (proved). Index attention reaches the map only through a global count, which needs two layers in principle but a normaliser, fine precision and large width (cited construction). Trained index models instead learn local return rules whose reach grows with depth in direction runs (analysis). Registered exchange rates: about three index layers on Dyck-2 and on a biased ring, more than four on the torus.
3. *A theory of when the map is found.* The clock/map decomposition and the bound it puts on one-pass retrieval; a route inventory for each step class (proved); a conjecture for why rank D starves the collective route in two or more dimensions but not in one. Registered support on both axes: the solution exists, is kept or re-found, and is not found from scratch at rank 2 (at most 2/8, also at twice the budget) or with a non-negative step (0/8 against 8/8, also at each step's best of four learning rates on a ring). Causal tests of the routes: freezing the move latents removes the collective route (1/8 against 7/8), and a non-negative readout with a signed increment restores it (16/16).

4. *Readers and codes.* Loops, extra layers and a score rule change how a code is read and never change the code family (proved). Loops and depth raise rank-2 accuracy (+0.079 and +0.096, registered), while the codes they learn stay far from rank 4's (analysis). PoPE's polar score pins every kernel's peak at zero phase difference (proved), which helps a path phase and hurts an index phase; freeing its phase offset raises the index model by +0.122 and leaves the path model at ceiling (registered).
5. *Maps from words.* One path layer learns navigation told in English (registered). On the revisits retrace cannot reach it scores 0.956 (0.978 on fresh seeds) against 0.411 for two index layers, below the 0.506 constant (analysis). Its step table follows one rule: a word's offset from its place depends on its role, not its identity, and a per-move clock survives only in channels retrieval does not use.

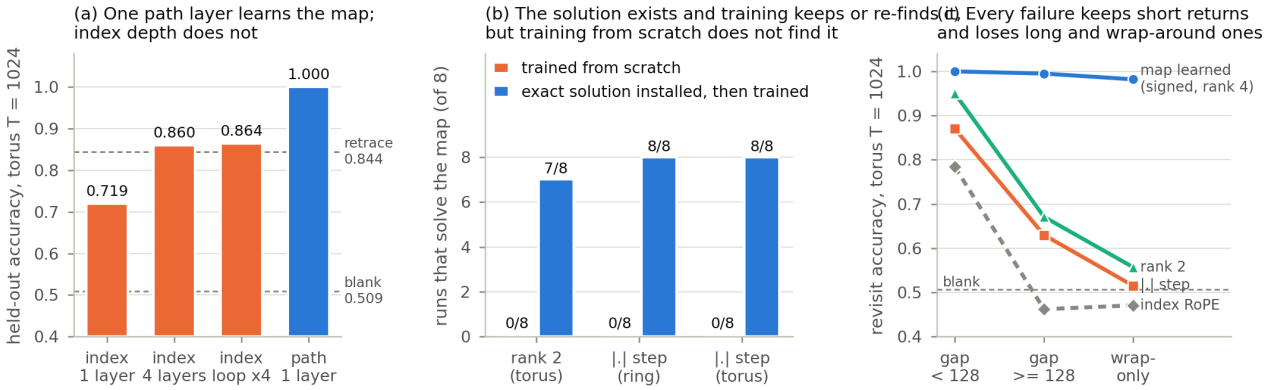


Figure 1 (`fig_summary`). The paper in three panels (torus, T = 1024 unless marked). (a) One path-integrated layer learns the map; index attention at one or four layers or looped four times does not (n = 4; dashed: this batch's retrace and always-blank floors, 0.844 and 0.509). (b) For both constrained classes a solution exists and training keeps it (rank 2: a rank-2 projection of a solved rank-4 model, 7/8 SOLVED after training; |.| on the ring: held 8/8) or re-finds it (|.| on the torus: perturbed early, 8/8 SOLVED at the end), while training from scratch solves 0/8. (c) Every failing one-layer model keeps short-gap revisits and loses long-gap and wrap-only ones; the model that learns the map holds all three (n = 8 per arm; dashed: always-blank within each stratum).

2. Background

2.1 Rotary position as an integrator. Write the phase of rotary channel c (a head and a two-dimensional block) at token t as theta_{c,t} = omega_c Phi_{c,t}, with Phi_{c,t} = sum_{u <= t} Delta_{c,u} the channel's summed step. The channel's contribution to the score of query t and key t' depends on position only through exp(-i omega_c (Phi_{c,t} - Phi_{c,t'})), the identity behind Mamba-3's Proposition 3 (App. E1). Howard et al. (2014) drive one bank of integrators by a rate alpha(t): a constant rate codes elapsed time (Case I), the speed codes distance travelled (Case II), and the signed velocity codes position, the same for every visit to a place, "like a bead on a wire" (Case III). The rotary families, as organised by GRAPE (Zhang et al. 2026), Selective RoPE and Mamba-3, sit in these three cases with an imaginary rate: index RoPE adds a constant angle per token; CoPE, CARoPE and non-negative content steps advance by a non-negative, content-dependent amount; MapFormer, Selective RoPE and Mamba-3's data-dependent RoPE add a signed

one, and alpha(t) is the selective-SSM step Delta_t. Velocity-modulated integrators have been trained with a real Laplace variable (Alipour et al. 2025), and RoPE has been derived as a velocity-controlled oscillator on given coordinates (GridPE, Li et al. 2024); the machine-learning models of the Howard and Tiganj labs keep the Laplace variable real (Mochizuki-Freeman et al. 2023; Kabir et al. 2025). In the rotary slot a signed rate is a bounded rotation, and the three cases differ not in what a channel can represent (Sec. 2.3) but in what the data and the optimiser make of it. TEM and TEM-t (Whittington et al. 2020, 2022) frame the Case III code as a "where" learned apart from the "what".

2.2 MapFormer. MapFormer's working-memory model (MapWM) rotates content queries and keys by theta_t = omega * cumsum(W_out W_in x_t), a low-rank content-to-angle map; observations learn near-zero steps and opposite actions opposite steps (their Fig. 3d). In their Table 2, one-layer MapWM with rank 2 solves 2D navigation at 128 steps, rank 1 does not, and RoPE, TAPE and CoPE with up to four layers fall short; they attribute CoPE's failure to the sign of its gate. We build on exactly this model. Our implementation reproduces their Table 2 2D result (0.989 for MapWM and 0.987 for MapEM, against 1.00 reported in v4 and 0.99 / 1.0 in v1; App. A4).

2.3 Representability. Real non-negative eigenvalues cannot track parity (Sarrof et al. 2024, Thm. 2; Grazzi et al. 2025, Thm. 1). Rotations escape this, and rotations by non-negative angles generate cyclic groups (Grazzi et al., Thm. 3; Mamba-3, Sec. 3.2). In the rotary slot the sign of a channel's step therefore does not limit what that channel can represent. Proposition 3 below extends this to our rank-limited step maps, so any difference between signed and non-negative steps in this paper is a difference in what training finds.

3. Setup and how map use is measured

Models. MapWM with one layer (up to four where stated), 2 heads, d_model 128, and step Delta = W_out W_in x through a rank-r latent. Unless stated, r = 4 and the latent is shared across heads; MapFormer's latent is per head, and per-head variants are named wherever rank is compared. *Per-head rank* is the rank of the map from token to the angles of one head. Three step parameterisations: *signed* (Delta = s = W_out W_in x), |.| (Delta = |s|, applied per channel after W_out) and *softplus* (Delta = softplus(s), so every step starts at ln 2). *Observation-step-free* variants give steps to the move tokens only. Each head has 32 two-dimensional rotary blocks (*channels*) with frequencies omega_i = 2 pi (1/Delta_max)^(i/31), Delta_max the grid size, as in MapFormer; omega is a free parameter whose sign is not constrained. The layer is otherwise a standard pre-norm transformer block (feed-forward width 4 d_model, dropout 0.1; about 204k parameters at one layer). The index baseline is RoPE with the canonical ladder base^(-2c/d), at 1-4 layers or as one block looped four times. We also pair PoPE (Gopalakrishnan et al. 2025), a rotary variant with a polar score, with its path-integrated version MapPoPE, and compare faithful CARoPE, CoPE and Selective RoPE (App. I).

Training and evaluation. AdamW (learning rate 1e-3, weight decay 0.05), 5% linear warm-up then cosine decay to 10% of the peak, 98 batches per epoch (16 walks per batch on the 1024-move torus, 128 on the ring and the 128-move torus), cross-entropy on revisited observations only (Dyck-2: learning rate 1e-4, next-token loss; App. A2). Torus

models are scored on 100 walks over one held-out map, so every torus number is transfer to a new map; ring models on 400 walks with freshly drawn maps.

Tasks. (i) *2D torus*: a 64 x 64 torus walked by runs of 1-10 repeated moves, interleaved with observations drawn from 16 objects. Walks have T = 1024 moves (2,048 tokens), 900 epochs; MapFormer's T = 128 is used for the floor analysis and the PoPE pair. (ii) *1D ring* of 32 cells, T = 128, forward-step probability p_plus (0.5 balanced; 1.0 never reverses), 300 epochs. (iii) *Dyck-2* at length 32 and nesting depth 12, scored by Hewitt et al.'s (2020) distance-averaged closing accuracy (chance 0.5). (iv) *Text world*: the torus walk rendered as English (58 words), T = 1024 words. (v) 2D and 3D tori of other sizes for rank against dimension. The loss and the accuracy are on *revisits*: observations at cells visited before, the only predictable targets.

Floors. For each task we measure the best map-free predictor: always-blank, the best n-gram over orders 0-5, and *retrace*: while the current run of moves undoes the previous run, copy the observation seen when the walk was last at the cell it is returning to. A score is evidence of a map when it clears these floors.

Strata. Revisits are split by what a map-free rule can reach. *Short-gap* revisits come fewer than 128 moves after the last visit to the cell; *long-gap* revisits come later; *wrap-only* revisits are to a cell seen before only from a different unwrapped position, a full lap of the torus away. Runs are at most 10 moves, so retrace never reaches a long-gap or wrap-only revisit; those two strata separate a map from a local rule. Figure 2 shows the strata for every model class in the paper, and we return to it throughout.

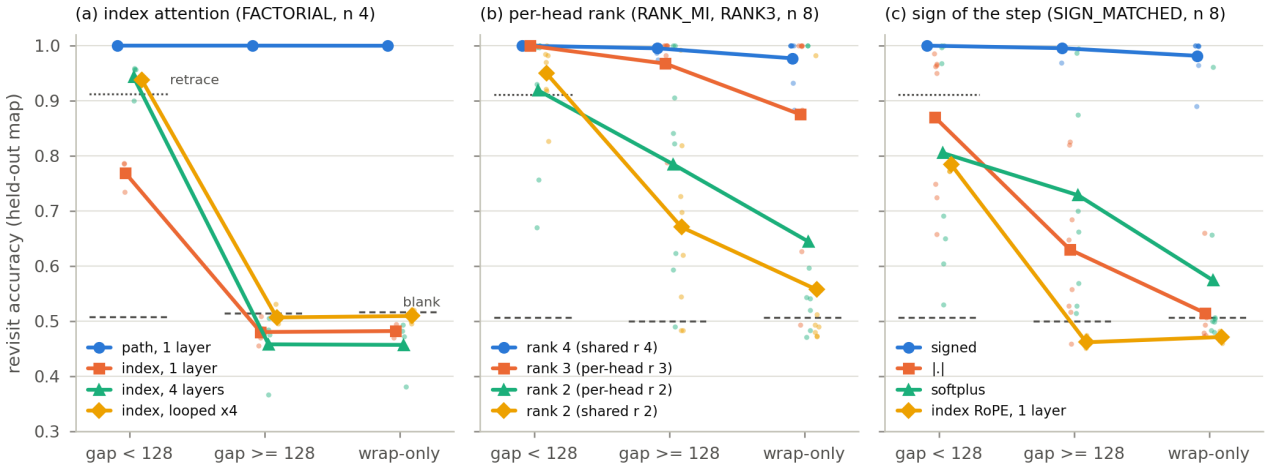


Figure 2 (`fig_strata_modes`). Revisit accuracy by stratum on the torus at T = 1024; lines: arm means, dots: runs. (a) Index attention (n = 4): depth and a loop help short-gap revisits only. (b) Per-head rank of the content-to-angle map (n = 8, same seeds and initial weights): rank 4 is near 1.000 in every stratum; rank 2 loses accuracy with the gap and most at wrap-only revisits. (c) Sign of the step (n = 8): signed near 1.000 everywhere; |.| and softplus fall from short gaps to wrap-only revisits; one-layer index RoPE for reference. Dashed: always-blank within each stratum; dotted: retrace within the short-gap stratum (0.91); floors on each batch's own walks.

Run classes and protocol. A run is *SOLVED* if its mean training loss over the final 5% of epochs is below 0.05; otherwise *STALLED* if its final-10% mean loss is within 5% of the preceding 10%, and *DESCENDING* otherwise. Every accuracy and SOLVED contrast

reported without the label "(analysis)" is a registered decision or a declared secondary of a pre-registered batch unless marked "not registered", with n = 8 seeds per arm unless stated, every model trained and tested at the same length and nesting depth, and exact permutation (accuracy) and Fisher (SOLVED) tests. Floors, strata floors and every number marked "(analysis)" were computed on stored streams and checkpoints, eval-only. App. A lists every batch with its seeds, its independent code check and its reproduction status; App. O lists every analysis script with its verification check.

4. Theory: maps and clocks in the phase

This section states what the results sections use; proofs are in App. E, and Table 1 gives the status of each statement. Notation: a channel c has angle a_c(x) = omega_c step_c(x) for token x; for a step map the latent of token x is z_x = W_in e_x and the channel's readout row is w_c, so step_c(x) = w_c . z_x before any |.| or softplus. The *usage* q_c of a channel is its mean query norm at move tokens times its mean key norm at observation tokens, normalised to sum to 1. A channel is *blind* to an inverse pair if both its angles are 0 mod 2 pi.

Proposition 1 (the prefix sum is in the phase; proved). The rotary score of channel c is Re(conj(q~*c) k~_c exp(-i omega_c (Phi*{c,t} - Phi_{c,s}))) with q~ the channel's query as a complex number. If every move token has a fixed step and every observation step is zero, theta_t = omega P_t, with P_t the unwrapped displacement after move t, and the score depends on position only through omega (P_t - P_s). One layer then compares two positions at any gap and any T, and wrapping costs nothing in a channel whose angle per move is a multiple of 2 pi / N.

Index attention sees position only through t - s, so "the moves in (s, t] sum to zero" must be computed in the residual stream. **Remark 1 (cited).** A global count gives it in two layers in principle: uniform causal attention averages the moves to P_t / n_t, an MLP rescales and takes the residue, and a second layer matches (Liu et al. 2023, Lemma 6; Yao et al. 2021). The construction needs a normaliser for the 1/n_t of causal averaging, precision 1/n_t (1/2048 per axis at the end of a torus walk), MLP width of order 4nT (64 x 2048 per axis here, against our 512), and a target whose sensitivity grows with the gap (Hahn and Rofin 2024, used for direction only). Rules that need no count, undoing the last run (retrace) or a fixed one-lap offset, are available to one layer. The measured exchange rates are therefore what training finds, not representational depth bounds.

Clock and map. For an inverse pair of moves (x, y) in channel c write the *clock* k_c = a_c(x) + a_c(y) and the *map* m_c = a_c(y). Over an interval with n_x moves x and n_y moves y, displacement d = n_y - n_x, the phase is m_c d + k_c n_x exactly. A *map code* has k_c = 0 mod 2 pi in every channel attention uses. To measure how far a trained code is from one we use a single ratio throughout the paper (Table 4, App. G3, H7): the *clock / map ratio* is the usage-weighted rms of wrap(k_c) over the usage-weighted rms of the map, with the map's size taken symmetrically over the pair, (wrap(a_c(x))^2 + wrap(a_c(y))^2) / 2, so that the readout does not depend on which move is called y; sums run over the inverse pairs and the channels of both heads (for one head, over its channels with its usage normalised to 1).

Proposition 2 (a bound on the position factor, and the tick rule; proved). (a) A zero-displacement revisit after n backward moves reads phase k_c n in channel c. If a usage share s of the channels has kappa <= |k_c| <= K, the same-cell position factor sum_c q_c cos(k_c n) is at most 1 - s (1 - cos(kappa n)) for n <= pi / K. The bound applies only while n |k| stays below pi: it says how fast a clock erodes the position match of a revisit, not that every clocked code fails at long gaps (by (b), a per-move tick of pi is invisible to revisits on the even torus). Index RoPE is the extreme case, in which every token advances every channel. (b) More generally, if the target is the key at the last visit to the same cell and returns come at lags with greatest common divisor g (g = 2 on the even torus), a channel can serve place-keyed retrieval only if its per-move *tick* is 0 mod 2 pi / g.

Proposition 3 (representability by wrapping; proved). If at most r tokens step, a |.| model of rank r reproduces any signed angle table exactly mod 2 pi; annihilating the observation embeddings in W_in leaves only the moves stepping, so every |.| model in this paper can represent the map. Softplus and exp can too: with the observations annihilated each steps softplus(0) = ln 2 or exp(0) = 1, and omega' = 2 pi / ln 2 or 2 pi makes that step a full turn.

Lemma 1 (relative-sign gauge; proved). Negating a channel's steps (through omega_c) together with the second coordinate of that channel's query and key projections leaves every attention score unchanged, since J R(phi) J = R(-phi) for J = diag(1, -1). A non-negative step therefore fixes only the relative signs of the tokens that step in a channel. For an inverse pair it removes cancellation through zero, angles (a, -a), while the wrapped code (a, 2 pi - a), the same function, remains.

Proposition 4 (route inventory; proved). (a) *Signed, collective route.* The clock vector is k = diag(omega) W u with u = z_x + z_y. The set {u = 0} cancels every channel at once and leaves the map free; gradient flow on a quadratic clock surrogate converges to it from every start; for almost every readout it is the only latent set that cancels every channel. (b) |.| *has neither route from a small start.* The same latent move ends at a(y) = a(x), a clock. At a generic readout no latent choice cancels more than 2r channels non-blindly. Per channel, an unwrapped |.| pair has |k_c| >= |m_c|, and its non-blind zero-clock points lie at |k_c| = 2 pi n, across |k_c| = pi from a small start. (c) *Rank.* Let c be the latent step common to all moves (with the mean observation step added when observations step) and U the map frame, the half-differences of opposite moves' latents. At per-head rank D, c lies in the plane U spans, c = U beta for a phantom displacement beta, so a channel is clock-free only by losing resolution along beta; above rank D the part of c outside the map plane can be cancelled channel by channel with the map untouched (*parking*). (d) *One dimension.* At D = 1 no readout change alters the drift-to-map ratio; only the collective route acts.

Proposition 5 (hiding; proved to first order) and Conjecture 1. At rank D >= 2 with isotropic resolution and a small drift, rotating a channel's resolution off the drift direction lowers the drift loss at zero first-order cost to the map (*hiding*). The collective route removes the drift at a rate proportional to the head's resolution along the drift direction, which hiding lowers. *Conjecture 1:* this feedback keeps the drift in place at rank D >= 2; in 1D there is nothing to hide along, which fits rank 1 learning the ring.

Remark 2 (exists, kept, not found). Let S be the solution set of a class and Bas the set of starting points from which training reaches S within the budget with probability at least one half. "A solution exists and an installed solution is kept, but training from scratch does not find it" says that S is non-empty, S lies in Bas, and the initialisation distribution puts little mass on Bas: the obstruction is where training starts, not what the class contains.

Proposition 6 (readers and codes; proved). In one layer, the query and key at a scored revisit depend on token identity only, so the model is a position-only reader of its own code. MapFormer computes theta once from the embeddings, so a loop or extra layers widen the reader and never change the code family (the set of codes the step map can produce). Any score of the form sum_c A_c cos(dphi_c + beta_c) splits target coherence into a code term C0 (the A-weighted cos dphi on the target) and a readout term C - C0; a score rule acts only on the second.

Proposition 7 (peak pinning; proved given delta = 0). PoPE's score sets beta_c = -delta_c with delta content-free and, in trained models, in its clamp's dead zone (delta = 0). Then every head's kernel, for every content pair, is a non-negative combination of cosines of dphi and peaks at dphi = 0. With a path phase dphi = 0 is the same place, the revisit target. With an index phase it is the current token, and with PoPE's frequencies (at most 1 rad per token) every content pair's kernel decreases over offsets 0-3, so no fixed far offset can be preferred in one layer.

Table 1. Status of the theory (proofs in App. E).

statement	status	checked against
Prop. 1, prefix sum in the phase	proved	same-cell attention of path models at long gaps (Sec. 6)
Remark 1, two-layer count	cited (Liu et al. Lemma 6; Yao et al.)	probes: no count in torus index models (Sec. 6)
Prop. 2, position-factor bound and tick rule	proved	bound: consistent with the strata of every failing model (Fig. 2), not a test of them; tick rule: text-world channels (Sec. 8)
Prop. 3 and Lemma 1, representability and gauge	proved	constructions exact to 1.4e-6 rad (App. D)
Prop. 4, route inventory	proved	end-state clocks (Sec. 7.1); frozen latents; hybrid readout (Sec. 7.3)
Prop. 5, hiding at first order	proved under small drift, isotropy, near-isometry	drift on the least-resolved axis (Sec. 7.2)
Conjecture 1, hiding starves the collective route	conjecture	consistent with rank 1 learning the ring; not tested
Remark 2, basins	definitional	existence and continuation batches (Secs. 7.2-7.3)
Prop. 6, readers and codes	proved	one-pass reader fits (Sec. 7.4)
Prop. 7, peak pinning	proved given delta = 0	delta-freedom batch (Sec. 7.4)
Prop. 8, CARoPE cannot cancel	proved	trained CARoPE models (Sec. 7.3)

5. One path layer solves the long-walk map

Short walks mostly do not need a map (analysis). On MapFormer's T = 128 torus walks retrace scores 0.934 (blank 0.507, n-gram 0.598), and no revisit is wrap-only. The walk moves in runs, and a run that reverses the previous one returns, step by step, to cells seen moments earlier; over 128 moves nearly every revisit is of this local kind. At T = 1024 retrace falls to 0.843 (blank 0.506, n-gram 0.575) and 7.9% of revisits are wrap-only (Table 2). At T = 128 the path model still beats index attention (path 0.971 at rank 2 and 1.000 at rank 4, index 0.805; position effect +0.243 at rank 2), but index RoPE falls below retrace there and the rank-2 path model clears it by only 0.04.

Table 2. Torus floors (analysis, on the same evaluation walks as the models) and accuracies at MapFormer's walk length and at ours (held-out map; n = 8 per model).

T (moves)	wrap-only revisits	blank	n-gram	retrace	index RoPE, 1 layer	path, rank 2	path, rank 4
128	0.000	0.507	0.598	0.934	0.805	0.971	1.000
1024	0.079	0.506	0.575	0.843	0.731	0.894	0.998

One path layer solves the long-walk map; index attention does not. On the torus at T = 1024 one path layer scores 0.998 +/- 0.004 (8/8 SOLVED) against 0.731 +/- 0.008 for index RoPE (0/8), a difference of +0.267 (permutation and Fisher p 0.0002). Index RoPE sits below retrace on every run (0.72-0.74; analysis): it does not learn even the map-free rule. Two further registered batches give the same contrast: 1.000 against 0.722 (n = 8) and 1.000 against 0.719 (n = 4; Fig. 3c). As Proposition 1 says, the path model's score depends on P_t - P_s directly, and it uses that at every gap.

Where the difference lives. By stratum (short gap / long gap / wrap-only), the path model scores 1.000 / 0.995 / 0.982 and one-layer index RoPE 0.784 / 0.462 / 0.471, against stratum floors of 0.506 / 0.500 / 0.507 and retrace 0.911 on short gaps (analysis) (Fig. 2c). Index attention gains its margin over the constant on short gaps, where retrace does better still; long-gap and wrap-only revisits require the map, and only the path model has it.

6. What index depth buys

Depth is the obvious way to give index attention more computation, so we ask how many index layers one path layer is worth, with both trained and tested at the same length and nesting depth.

Dyck-2: about three index layers. A bracket string is a walk on a stack: an opening bracket steps forward, its closing partner steps back, and MapFormer's models encode the two as opposite directions. At length 32 and nesting depth 12 one path layer scores 0.996 (1.000 at 2-4 layers), and index attention 0.648 / 0.889 / 0.964 / 0.982 at 1-4 layers, against a best stack-free floor of 0.594 (Fig. 3a). The position main effect, averaged over two encodings (MapWM vs RoPE and MapPoPE vs PoPE; App. F), is +0.353 / +0.130 / +0.045 / +0.024 at 1-4 layers, positive on 8/8 seeds at each depth (sign-flip p 0.0078). With three times the budget four index layers reach the ceiling (0.997-0.998), and training on a mixture of depths 4-12 keeps an effect of +0.110 at depth 12 (8/8). The

index arms are not held back by their frequency ladder: base 32 and base 10000 agree to 0.001 at four layers.

A biased 1D walk: the same rate. The ring dials how often a step is undone: at p_plus 0.5 the walk is balanced, at 0.9 one segment in ten runs backwards, and at 1.0 the walk never reverses and position is a count. On fresh seeds (n = 6) one path layer scores 1.000 at every bias, and index attention with 1 / 2 / 3 layers 0.710 / 0.949 / 0.998 at p_plus 0.5, 0.677 / 0.962 / 0.998 at 0.75 and 0.824 / 0.987 / 0.999 at 0.9 (Fig. 3b; declared secondary). Wherever a step can be undone, three index layers come within 0.002 of the path layer; at p_plus 1.0 a single index layer reaches 1.000.

The 2D torus: more than four. At T = 1024 (n = 4) one path layer scores 1.000 (4/4 SOLVED) against index attention at one layer 0.719, at four real layers 0.860 and as one block looped four times 0.864 (each 0/4; complete separations, p 0.029) (Fig. 3c). Depth and the loop each help (+0.140 and +0.145), and the loop matches four real layers with a quarter of the parameters (203,925 against 798,741). Four layers or a loop lift short-gap revisits from 0.769 to 0.943 / 0.938 and leave long-gap (0.458 / 0.507) and wrap-only (0.457 / 0.509) revisits at the constant floor (0.515 / 0.517; Fig. 2a).

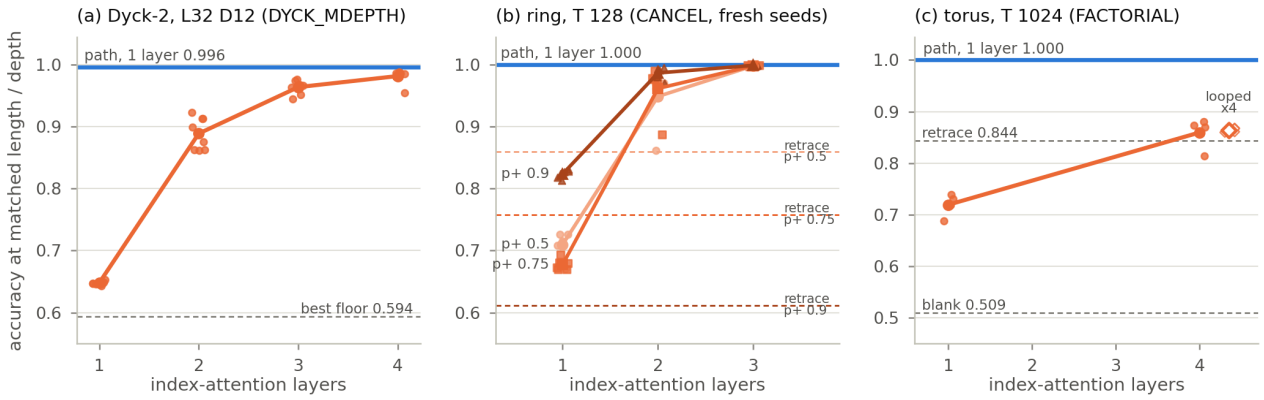


Figure 3 (`fig_exchange_rate`). Accuracy of index-RoPE attention against its number of layers (orange; dots: runs, line: means) beside one layer of path integration (blue line), each trained and tested at the same length and depth. (a) Dyck-2 at length 32 and depth 12 (Hewitt closing accuracy; n = 8): index attention needs about three layers to approach one path layer (0.964 at 3, 0.982 at 4, against 0.996). (b) Ring of 32 cells at T = 128 for three move biases (n = 6 fresh seeds; declared secondary): three index layers reach one path layer (0.998-0.999 against 1.000). (c) Torus at T = 1024 (n = 4): four index layers (0.860) or one layer looped four times (0.864) stay far below one path layer (1.000), just above this batch's retrace floor (0.844; the 0.843 of Table 2 is the floor on the sign batch's walks). Dashed lines: map-free floors (analysis; Dyck: the best stack-free floor; ring: retrace per bias, on another stream of the same task; torus: always-blank and retrace on this batch's own walks).

Why the rate differs by task. A revisit asks for an earlier observation at the same cell, that is, for a key whose moves since then sum to zero. Remark 1 offers index attention a global count or rules that need none. The stored models show that training finds only the second kind, with a reach that differs by task (analysis; Table 3, App. F). One index layer learns retrace and the one-lap offset: on the ring it scores 0.68-0.80 on retraces, 0.99-1.00 on one-lap returns once they are common, and the floor on everything else at

p_plus 0.5 and 0.75 (0.487 against 0.491-0.493; at 0.9 an unexplained 0.592 against 0.502), so at p_plus 1.0, where every revisit is a one-lap return, it matches the path layer. Three index layers on the ring are flat in the number of direction runs a revisit spans (0.997 / 0.993 / 0.972 at 3-4 / 5-8 / 9-16 runs, wrap-only revisits included; 0.999 / 0.996 / 0.988 on plain revisits only, Fig. 4b) and carry a displacement code over the last ~64 moves (linear-probe R^2 0.72 for the displacement over the last 64 moves, read at moves 96-127); displacement older than 64 moves is not decodable (R^2 0.01), so no global count is detectable on the ring either, and on 128-move walks a 64-move window covers most revisits. On the torus no index model carries a count (R^2 at most 0.21). Four layers or the loop solve retraces (0.996 / 1.000) and decay with the number of runs spanned, at a fixed gap as well as overall (Fig. 4a): at gaps of 32-127 moves four layers score 0.874 at 2 runs and 0.493 at 17 or more (seed means). Their last layer puts 0.55-0.58 of its attention on the remembered cell for retraces and 0.00-0.01 at long gaps, where the path layer puts 0.81. Nearly all long-gap revisits span 17 or more runs (98.5%; median 37), beyond that reach. Short-gap accuracy decomposes over these classes: the retrace floor is (0.680 x 1 + 0.148 x 0.5) / 0.828 = 0.911, and four layers give (0.680 x 0.996 + 0.148 x 0.699) / 0.828 = 0.943, so depth adds the exact retrace rule plus partial returns over 3-8 runs.

Table 3. Torus at T = 1024: accuracy by revisit class (analysis; FACTORIAL models, n = 4, eval mode; "runs" = maximal runs of one move between the two visits; share = fraction of all revisits; class contrasts at n = 4 carry no test).

revisit class	share	index 1 layer	index 4 layers	index loop x4	path 1 layer
short gap, retrace	0.680	0.833	0.996	1.000	1.000
short gap, not retrace: 3-4 runs	0.031	0.477	0.926	0.977	1.000
... 5-8 runs	0.056	0.476	0.735	0.645	1.000
... 9-16 runs	0.052	0.474	0.561	0.499	1.000
... 17+ runs	0.009	0.472	0.493	0.502	1.000
long gap (>= 128 moves)	0.090	0.480	0.458	0.507	1.000
wrap-only	0.083	0.482	0.457	0.509	1.000

So depth buys index attention a local reach measured in direction runs, and the torus rate is "more than four" because its long-range revisits lie beyond that reach and the global count is not found. The same four passes do something else on a rank-2 path code: they lift long-gap revisits from 0.671 to 0.927-0.990 and wrap-only revisits from 0.557 to 0.747-0.887 (Sec. 7.4). Extra passes add accuracy where the code carries position information that one pass cannot read, and none where it carries none.

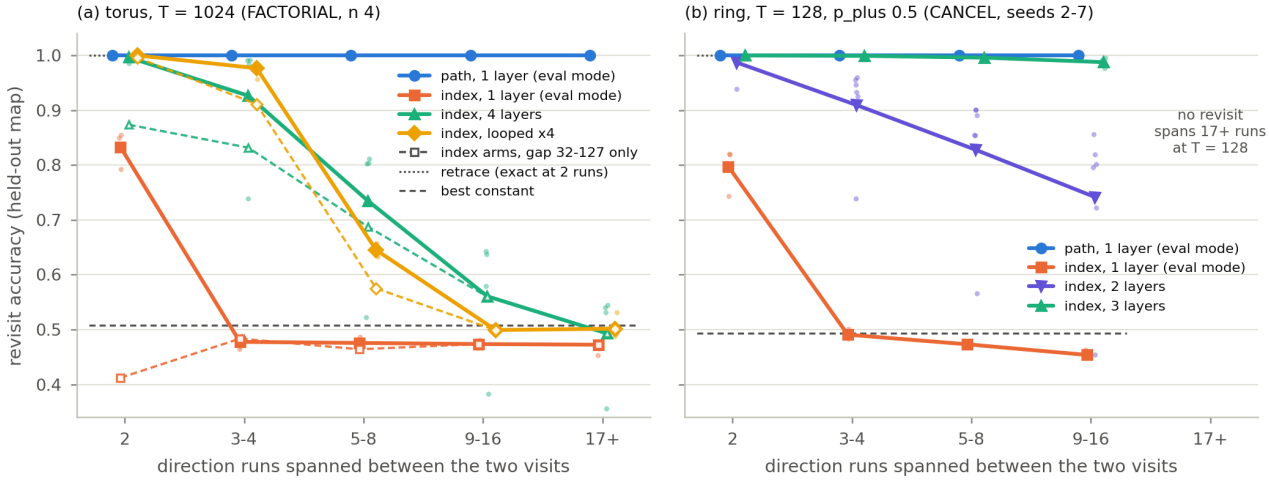


Figure 4 (`fig_segments`; analysis of registered batches, eval mode). Revisit accuracy against the number of direction runs (maximal runs of one move) the walk spans between the two visits; lines: means of per-seed accuracies (seed means, not pooled); dots: seeds. (a) Torus at T = 1024 (FACTORIAL, n = 4). Solid: plain revisits at gaps under 128 moves (2 runs = retrace; the rows of Table 3). Four index layers fall from 0.996 at 2 runs to 0.926 / 0.735 / 0.561 / 0.493 at 3-4 / 5-8 / 9-16 / 17+ runs and the loop from 1.000 to 0.977 / 0.645 / 0.499 / 0.502; one index layer (eval mode) scores 0.833 on retraces and 0.472-0.477 beyond; one path layer scores 1.000 in every bin. Dashed: plain revisits at gaps of 32-127 moves only, the fixed-gap control: the same decay (four layers 0.874 at 2 runs to 0.493 at 17+, the loop 0.996 to 0.502). (b) Ring of 32 cells at T = 128, p_plus 0.5 (fresh seeds 2-7, n = 6), plain revisits only (wrap-only revisits and one-lap returns excluded; 2 runs = retrace): three index layers stay flat (1.000 / 0.999 / 0.996 / 0.988 at 2 / 3-4 / 5-8 / 9-16 runs), two layers decay (0.987 / 0.910 / 0.828 / 0.742), and one layer (eval mode) learns retrace only (0.797, then 0.454-0.491); no ring revisit spans 17 or more runs. Floors on each batch's own walks: retrace is exact on 2-run revisits (dotted) and falls to the best constant elsewhere (dashed: the torus short-gap stratum's 0.508; the ring's 'neither' class 0.493, a class that includes wrap-only revisits).

7. What decides whether training finds the map

Two choices of the content-to-angle map decide whether the torus map is learned: its per-head rank and the sign parameterisation of its step. For each, the constrained class fails from scratch, a solution in the class exists and is constructed explicitly, and that solution, installed and trained, is kept or re-found. Section 4 says what to look for: whether training removes the clock, and by which route.

7.1 The clock and the signature

Training from a small start ends in one of two places on either axis (Table 4, Fig. 5; analysis; the clock / map ratio of Sec. 4). The models that succeed end with clock-free heads: the signed and rank-3/4 arms at arm means 0.001-0.051, with no head above 0.14, and every solved rank-2 run with at least one clean head (0.03-0.07; two of the four keep a drifting second head, 0.54 and 0.68). The models that fail end with a clock of the order of their map: the non-negative arms at 1.25-1.57 (arm means; for an unwrapped |.| channel a ratio of at least 1 holds by construction, so the informative part is that the

channels did not wrap) and the unsolved rank-2 runs at 0.54-0.69 (medians 0.58-0.79). The exceptions are informative too: 4 of the 20 unsolved rank-2 runs end with clock-free heads whose two map axes have collapsed onto one direction (a map that has lost a dimension; App. G3), and one more sits just above the SOLVED threshold (final loss 0.051). The measured strata show the signature that Proposition 2's bound leads one to expect: shared rank 2 scores 0.950 / 0.671 / 0.557 on short-gap / long-gap / wrap-only revisits and |.| 0.870 / 0.630 / 0.515, against 1.000 / 0.995 / 0.982 for the signed rank-4 model (Fig. 2b-c). The bound alone does not predict failure: two signed torus runs started with angle sums past pi solve with clock / map 1.39 and 1.78, with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi, where the bound says nothing (App. H7). Weighted by usage, signed steps cancel opposite moves mod 2 pi in 70-93% of channels and |.| steps in 2-10% (analysis; App. H6).

Table 4. Clock at the end of training (analysis; clock / map ratio of Sec. 4, one value per run, arm means unless marked; SOLVED of 8). For unwrapped |.| channels the ratio is at least 1 by construction; the informative rows are the frozen-latent signed arm and the rank arms.

arm	SOLVED	clock / map
ring, signed (three batches)	8/8, 7/8, 8/8	0.015-0.018
ring, |.| from small starts (three batches)	0/8, 0/8, 1/8	1.47-1.57
ring, signed with move latents frozen, |.| start	1/8	1.18 (the solved run 0.06)
ring, |.| from a start past pi	8/8	0.32 (wrapped; 0.82 of usage on cancelling channels)
torus, signed / |.|, observation-step-free	8/8 / 1/8	0.001 / 1.25
torus, rank 2, unsolved runs (three arms)	0/8, 2/8, 2/8	0.54-0.69 (medians 0.58-0.79); 4 of 20 runs collapsed, clock-free
torus, rank 2, solved runs (4): six clean heads / two drifting heads (per head)	4 runs	0.03-0.07 / 0.54, 0.68
torus, rank 3 and rank 4, solved runs	6/8, 8/8, 8/8	0.043-0.051

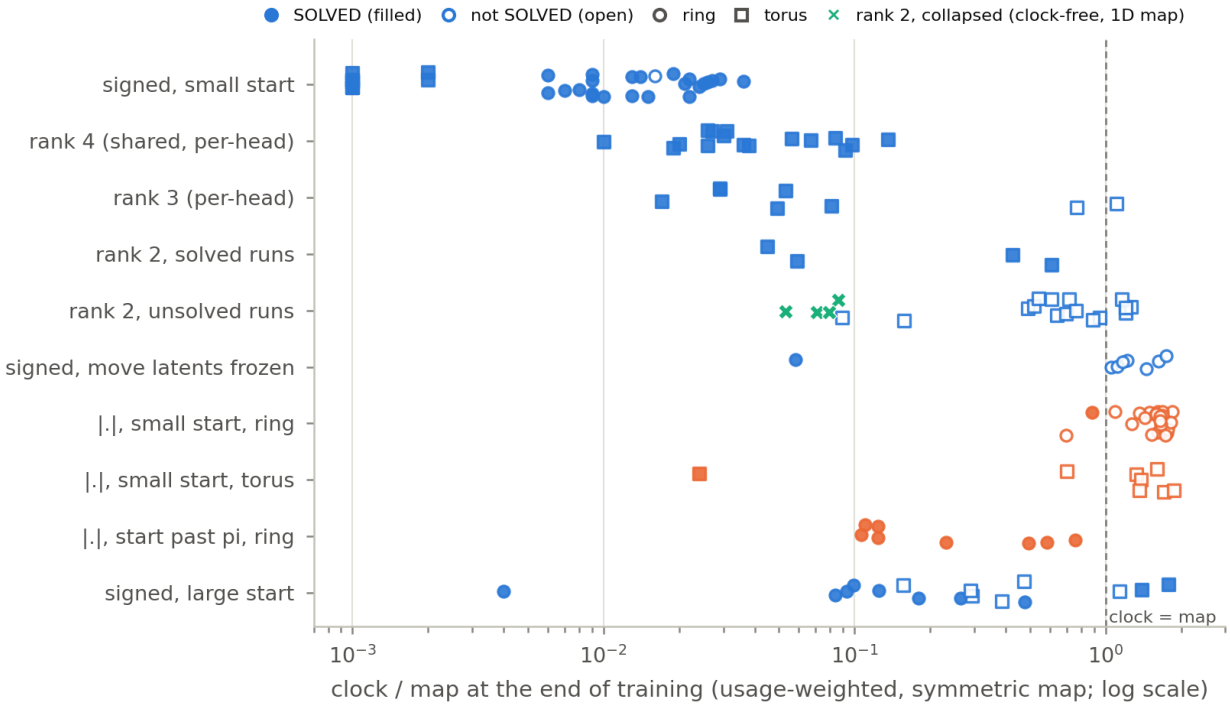


Figure 5 (`fig_clock_endstate`; analysis of registered batches). Clock / map ratio at the end of training, one dot per run (circles: ring; squares: torus; filled: SOLVED; log scale), on the one definition of Table 4 throughout (symmetric map, usage-weighted over both heads). Successes from small starts are clock-free: signed 0.001-0.036 (31 of 32 runs SOLVED; the unsolved one at 0.016), rank 4 0.010-0.136 (16/16), solved rank 3 0.017-0.081. Failures carry a clock of the order of the map: |.| from small starts 0.70-1.85 on the ring (1 of 24 SOLVED, at 0.88) and 0.70-1.87 for the unsolved torus runs (the solved one at 0.024), signed with move latents frozen 1.055-1.741 (the solved run 0.058), unsolved rank 3 0.767-1.105 and unsolved rank 2 up to 1.26 (median 0.62). Four unsolved rank-2 runs are clock-free (0.053-0.086) with their two map axes collapsed onto one direction (crosses; App. G3). The solved rank-2 runs span 0.045-0.611 (two keep a drifting second head). Starts past pi cancel by wrapping and solve at larger ratios: |.| on the ring 0.11-0.76 (8/8); signed 0.004-1.78 (10 of 16, ring and torus), the two solved torus runs at 1.39 and 1.78 with 0.43 and 0.99 of their usage on channels whose per-pair clock is near pi. For an unwrapped |.| channel a ratio of at least 1 holds by construction.

7.2 Per-head rank of the content-to-angle map

Per-head rank decides it. On the torus at T = 1024 we compare six bottlenecks, all built from the same initial weights at each seed (Table 5, Fig. 6a). Every arm whose heads each read a rank-2 map solves 0-2 of 8; rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. At matched sharing (per-head rank 4 against a block-diagonal rank-4 latent in which each head reads rank 2) the split is 8/8 against 2/8 (Fisher and permutation p 0.0070). Per-head rank 3 improves on per-head rank 2 by +0.102 in accuracy (permutation p 0.027, Holm 0.054). Total latent size does not track success: the per-head rank-2 arm and the shared rank-4 arm both have four latent dimensions and land on opposite sides. Nor does the initial angle scale: two rank-2 arms start at the same scale as rank 4 and fail. Most rank-2 runs flatten at losses of 0.2-0.5, while the rank-3 and rank-4 medians cross the SOLVED threshold at epochs 478-532

(App. G). With MapFormer's Table 2, the picture is: rank 1 cannot represent a 2D map; rank 2 can, and learns 128-step walks; at 1024 moves reliable learning needs per-head rank above the map's dimension.

Table 5. SOLVED (of 8) and held-out accuracy at T = 1024 (mean over 8 seeds) by per-head rank; torus, 900 epochs, every arm built from the same initial weights at each seed.

per-head rank	arm	SOLVED	accuracy
2	one rank-2 latent shared by the heads	0/8	0.894
2	per-head rank-2 latents	2/8 (1/6 on seeds 2-7)	0.885
2	block-diagonal rank 4 (each head reads rank 2)	2/8	0.948
3	per-head rank 3	6/8	0.987
4	shared rank 4	8/8	0.998
4	per-head rank 4	8/8	0.999

Twice the budget does not rescue rank 2. Trained for 1800 epochs from scratch, rank 2 solves 0/8 (0.908) against rank 4's 7/8 (Fisher p 0.0014; accuracy +0.086, permutation p 0.0003). PoPE's polar score does not rescue it either: 3/12 SOLVED under either score at T = 1024, with rank 4 at 8/8 under both (Sec. 7.4 gives the reason).

The rank-2 solution exists and is held. The rank-2 projection of each solved rank-4 model scores 0.9955 frozen (at least 0.981 on every seed). Trained from there it solves 7/8, as many as the rank-4 control trained from its own solution (7/8), against 1/8 for rank 2 trained further from its own stalled weights (accuracy p 0.0009) (Fig. 7a). A rank-2 map that solves the torus exists, is kept by training, and is not reached from scratch.

Rank against the map's dimension. Rank equal to the map's dimension D is hard in 2D and 3D and not in 1D (Fig. 6b). On a 2D torus of 32 cells per side rank 2 solves 1/8 against rank 3's 8/8 (Fisher p 0.0014). On a 3D torus of 10 cells per side rank 3 solves 1/8 (rank 4, 4/8), and rank 4 solves an 18-per-side 3D torus 8/8 (1.000). On the 1D ring rank 1 solves 7/8 against rank 2's 8/8, and with the drift forbidden by construction rank 1 solves 8/8 in a median of 24 epochs instead of 58.

Why the extra rank helps. Proposition 4(c) gives a channel above rank D a map-free way to stop reading the clock. The stored checkpoints are consistent with it but do not discriminate it from plain removal of the clock through the latent (analysis; App. G4). Here the clock is c = a + o, the mean move latent a plus the mean observation latent o, the drift a walk accumulates per move and observation. In solved heads of rank above D, a and o are each 0.02-0.04 of a map step and point in opposite directions (cosine -0.79 to -0.999), so they cancel; at T = 1024 both lie mostly in the map plane. What is left of c (0.001-0.025 of a map step) lies almost entirely outside the plane: off-plane share 0.96-1.00 in nine of ten sets of heads, against 0.40-0.71 for a random direction (median), read at 0.09-0.27 of the map's gain. Parking and removal through the latent would both leave such a residual. Drifting rank-D heads drift along their least-resolved axis (50 of 55 runs, counting each network once, against 25.1 expected for a random direction), the geometry Proposition 5 leads to. A fresh rank-4 step map trained on the stalled rank-2 networks removes the drift in 16 of 16 heads: its latent clock shrinks 14-85x in absolute size, and the residual (0.002-0.005 of a map step) lies off the plane (share 0.91-1.00). At

rank 1 on the ring, Proposition 4(d) leaves only the collective route, and Conjecture 1 has nothing to starve: rank 1 learns.

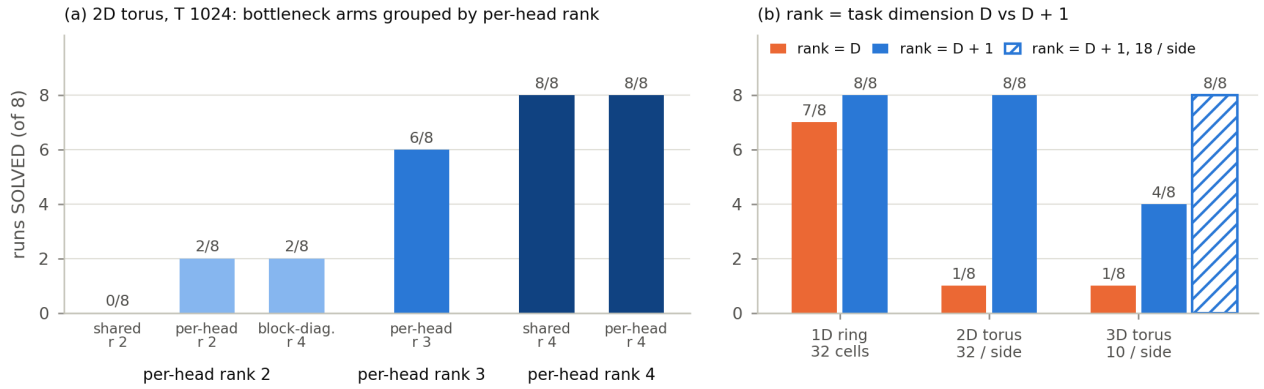


Figure 6 (`fig_rank_solved`). Runs (of 8) that solve the map, by the per-head rank of the content-to-angle map. (a) Torus, T = 1024, six bottlenecks built from the same initial weights: every arm whose heads each read a rank-2 map solves 0-2/8, rank 3 solves 6/8 and rank 4 8/8, whether the latent is shared or per head. (b) Rank equal to the task's dimension D against D + 1 (separate batches and recipes): on the 1D ring rank 1 solves 7/8 (rank 2, 8/8); on a 2D torus rank 2 solves 1/8 (rank 3, 8/8); on a small 3D torus (10 per side) rank 3 solves 1/8 and rank 4 4/8, and rank 4 solves 8/8 on an 18-per-side torus with few wrap-only revisits (hatched). Torus budget 900 epochs; at 1800 epochs rank 2 still solves 0/8.

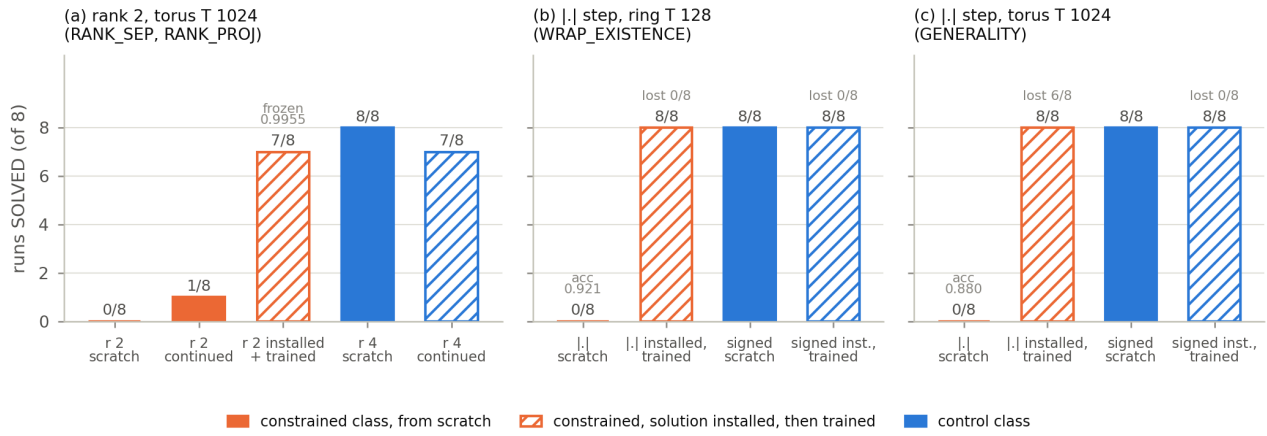


Figure 7 (`fig_existence`). A solution exists in each constrained class and training keeps or re-finds it, but training from scratch does not find it. Bars: runs SOLVED of 8 (hatched: the solution was installed and then trained). (a) Rank 2 on the torus: a rank-2 projection of a solved rank-4 model scores 0.9955 frozen and is held when trained (7/8), while rank 2 from scratch solves 0/8, and 1/8 when the stalled runs are trained further. (b) The |.| step on the ring: an installed exact non-negative solution is held under training (8/8 SOLVED, lost on 0/8), while |.| from scratch solves 0/8 against signed 8/8. (c) The |.| step on the torus: the installed solution is perturbed (loss above 0.05 on 6/8) and re-found (8/8 SOLVED); from scratch 0/8 against 8/8. "Lost": the maximum per-epoch training loss after the start exceeds 0.3, or exceeds 10x the signed control's maximum at that seed and 0.05.

7.3 The sign of the phase step

Non-negative steps rarely learn the map from scratch. On the torus at T = 1024 the |.| step scores 0.821 against 0.998 for the signed step, -0.177 (permutation p 0.0002), and solves 0/8 against 8/8; softplus scores 0.781 and solves 1/8 (-0.218, p 0.0003). On the ring |.| solves 2/8 and 1/8 against 8/8 at p_plus 0.5 and 0.75 (Fisher p 0.007 and 0.0014). Observation-step-free models show the same: 0/8 against 8/8 on the ring and on the torus (0.880 against 1.000). The training curves separate (Fig. 8): the signed median crosses the SOLVED threshold at epoch 480 / 57 / 309 / 183 in four batches and keeps descending, while most |.| runs flatten at a loss of 0.3-0.5.

The cost holds at each step's best learning rate, without weight decay and at 2 layers d_model 256 (Table 6). With each step at its own best of four learning rates on the ring (the |.| optimum, 3e-3, is bracketed by the grid), |.| solves 2/8 against signed 8/8 (selection-aware p 0.0070 on SOLVED and 0.0057 on accuracy). Without weight decay the ring gives 0/8 against 8/8 (-0.127); at a third of the learning rate, 0/8 against 6/8. A 2-layer, d_model 256 torus model with eight times the parameters gives 2/8 against 8/8 (0.967 against 0.999, p 0.0005). Across the learning-rate grid, higher rates hurt the signed step first (8, 8, 4 and 1 of 8 solved at 1e-3, 3e-3, 1e-2 and 3e-2) and do not help |.| past 3e-3 (1, 2, 1 and 0 of 8). Two further non-negative increments behave alike at the two reference rates: exp, the LRU parameterisation, and softplus, both of which start every token at the same positive step, solve 1/8 to 3/8 against signed 8/8 (App. H).

Table 6. The sign cost by task, model and recipe (observation-step-free models unless marked "full"; n = 8 per arm).

task and model	recipe	|.| accuracy, SOLVED	signed accuracy, SOLVED	test
torus, 1 layer, full	default (lr 1e-3, weight decay 0.05)	0.821, 0/8	0.998, 8/8	perm p 0.0002; Fisher 0.0002
torus, 1 layer	default	0.880, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
torus, 2 layers, d_model 256	default	0.967, 2/8	0.999, 8/8	perm p 0.0005; Fisher 0.0070
ring, full, p_plus 0.5	default	0.973, 2/8	1.000, 8/8	Fisher 0.0070
ring	default	0.911, 1/8	1.000, 8/8	perm p 0.0002; Fisher 0.0014
ring	weight decay 0	0.873, 0/8	1.000, 8/8	perm p 0.0002; Fisher 0.0002
ring	lr / 3 (3.3e-4)	0.841, 0/8	0.950, 6/8	Fisher 0.0070
ring	each step at its best of four rates (|.| 3e-3, signed 1e-3)	0.963, 2/8	1.000, 8/8	selection-aware p 0.0070 (SOLVED), 0.0057 (accuracy)

The non-negative solution exists, and training keeps or re-finds it. Proposition 3 constructs it: each solved signed observation-step-free model is converted exactly into |.| form, with the move latents re-solved to orthogonal directions, each channel's frequency

set to max(|omega|, 1) and each step to the signed angle reduced mod 2 pi. Angles are unchanged mod 2 pi (error at most 1.4e-6 rad), frozen accuracy equals the source's on 8/8 seeds, and the weights stay within about 5x of the signed model's, so the test is posed at the scale training uses (App. D). On the ring the constructed |.| model, trained for 300 more epochs, stays solved on 8/8 seeds and never leaves the solved region, keeping its cancelling code in the channels attention uses (98%, usage-weighted). On the torus, trained at learning rate 1e-4, it is perturbed early (peak loss 0.11-0.21), re-enters the solved region for good between epochs 119 and 204, and ends solved on 8/8 seeds at accuracy 1.000. From scratch the same class solves 0/8 on both tasks (Fig. 7b-c).

Why the non-negative step does not find it. Proposition 4(b) says the |.| class lacks both routes from a small start, and the stored models show the consequences (analysis; App. H). At initialisation a latent move produces exactly the change of angle sums the signed step needs (share 1.000) but only 0.34 of what |.| needs (0.56 usage-weighted); the rest must come from per-channel readout changes. With the content of a solved model held fixed, the loss along the straight path from the start to the |.| solution rises above its running minimum by 0.6-1.9 nats on every seed, against 0.04-0.48 for the signed path (complete separation). Rotating one move latent toward the negative of the other, with nothing else changed, takes the signed model from 0.30 to 0.72 accuracy and leaves the |.| model at 0.30, since under |.| that move makes a clock. The cone of Proposition 4(b) is where trained |.| ring models end: 0.25-0.36 of their sub-pi channels lie within 10% of its boundary, where one move's step is zero, against 0.056 at initialisation. In full training the content co-adapts along the collective route; with the content frozen, the step map alone rarely takes it from the same-direction start (exploratory: these CPU runs are not deterministic, and single runs moved by up to 0.43 in accuracy on re-runs; App. H7).

The cost is carried by the increment, not the readout. A hybrid whose readout matrix is confined to the non-negative orthant, a_c(x) = omega_c |w_c| . z_x, but whose latent and hence increment are signed, keeps the collective route. On the ring it solves 8/8 at both reference rates, at accuracy 1.000 on every seed, exactly as signed, against |.| 0/8 and 2/8 (Fisher p 0.0002 and 0.007), and ends where Proposition 4(a) says: antipodal move latents (median |z_x + z_y| / (|z_x| + |z_y|) 0.043 and 0.002; cosine -0.996 and -1.000). A |.| control with the move latents tied antipodal by construction, whose two steps are then equal and whose code is a pure clock, solves 0/16. Non-negativity in the readout costs nothing; a non-negative increment, which removes both the collective route and the near-side cancelling point, is what costs.

How signed steps reach the map. Started from the |.| start, with both moves rotating the same way, signed steps still solve 7/8 on the ring (|.| from the same start: 0/8; Fisher p 0.0014). They end with 98-99% of channels opposite-signed (analysis): they cancel opposite moves through zero, which Lemma 1 shows the non-negative step excludes for an inverse pair (Fig. 9). The escape is collective. Freezing the two move latents, with W_out and omega still free, blocks it: 1/8 against 7/8 (Fisher p 0.010), and the frozen-latent model ends where the non-negative step ends (clock / map 1.18, Table 4). With the latents free, one move's latent moves toward the negative of the other's and every channel cancels together (82% of channels end opposite-signed, against 15% with the latents frozen). MapFormer described this antipodal end state (their Figs. 3d and 9b); this is the causal test of how it is reached. A |.| step has only the wrapped branch, and on

the ring it learns the map when its opposite-move angles start with sums past pi (8/8, cancelling by wrapping; Fig. 9d).

Published content phases also miss the torus map. Faithful CARoPE cannot represent cancellation. *Proposition 8 (proved):* CARoPE advances every channel by an amount in (0, 1] per token, so the shortest return's phase lies in (0, 4], never 0 mod 2 pi, and every channel is either a clock or blind over an excursion (App. E5). It scores 0.667 on the torus (0/8), below index RoPE's 0.722 (p 0.040). Faithful CoPE can represent ring cancellation (we construct the solution; App. I) yet solves 0/8 on the ring (0.887 against signed 1.000, p 0.0002) while beating index RoPE by +0.157, and it fails a 256-cell line that never wraps (1/16 against signed 8/8), so its obstacle is not what it can represent. Faithful Selective RoPE, a signed content-dependent phase in its authors' best language-model configuration, builds a phase it relies on (it beats index RoPE by +0.236, and zeroing its angle drops it from 0.961 to 0.507) but does not learn the torus map within 900 epochs: 0/8 SOLVED against MapWM's 8/8 in the same batch (0.961 against 1.000, p 0.0002), with every run still descending. Its readouts show why the map is harder for it: every token steps the phase, and an observation steps 0.86-0.96 as far as a move (MapWM: 0.027), so what is seen moves where the model thinks it is. Its state-tracking configuration carries a positive bias that keeps every learned step non-negative, and it fails as the non-negative steps do (0.769, 0/8).

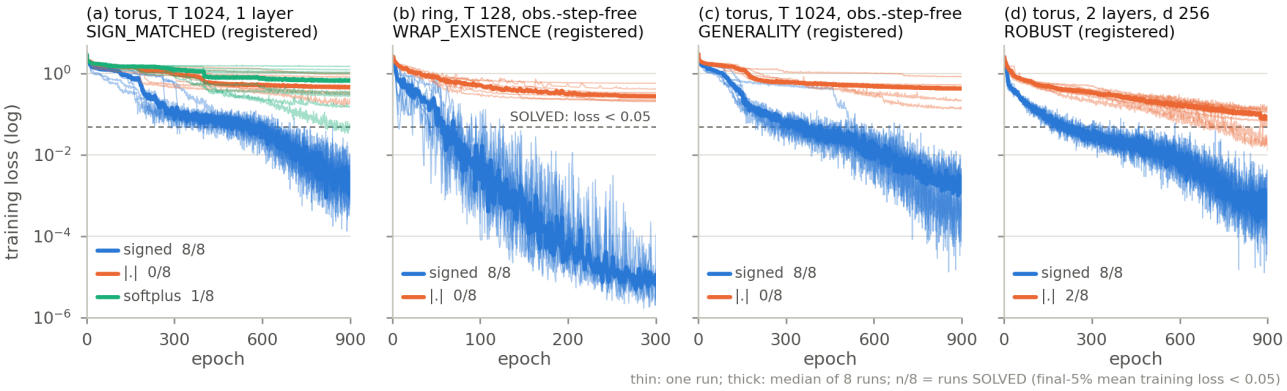


Figure 8 (`fig_loss_curves`). Training loss per epoch for the signed step (blue) and the non-negative steps |.| (orange) and softplus (green), every run of four batches (thin lines; thick: median of 8 runs; log scale; training loss with dropout on). Dashed line: the SOLVED threshold (final-5% mean training loss below 0.05); n/8 = runs SOLVED. (a) Torus, T = 1024, one layer: signed 8/8, |.| 0/8, softplus 1/8. (b) Ring of 32 cells, observation-step-free: 8/8 against 0/8. (c) Torus, observation-step-free: 8/8 against 0/8. (d) As (c) at two layers and d_model 256: 8/8 against 2/8. The signed median crosses the threshold at epoch 480 / 57 / 309 / 183 of 900 / 300 / 900 / 900 and keeps descending; the |.| median never does.

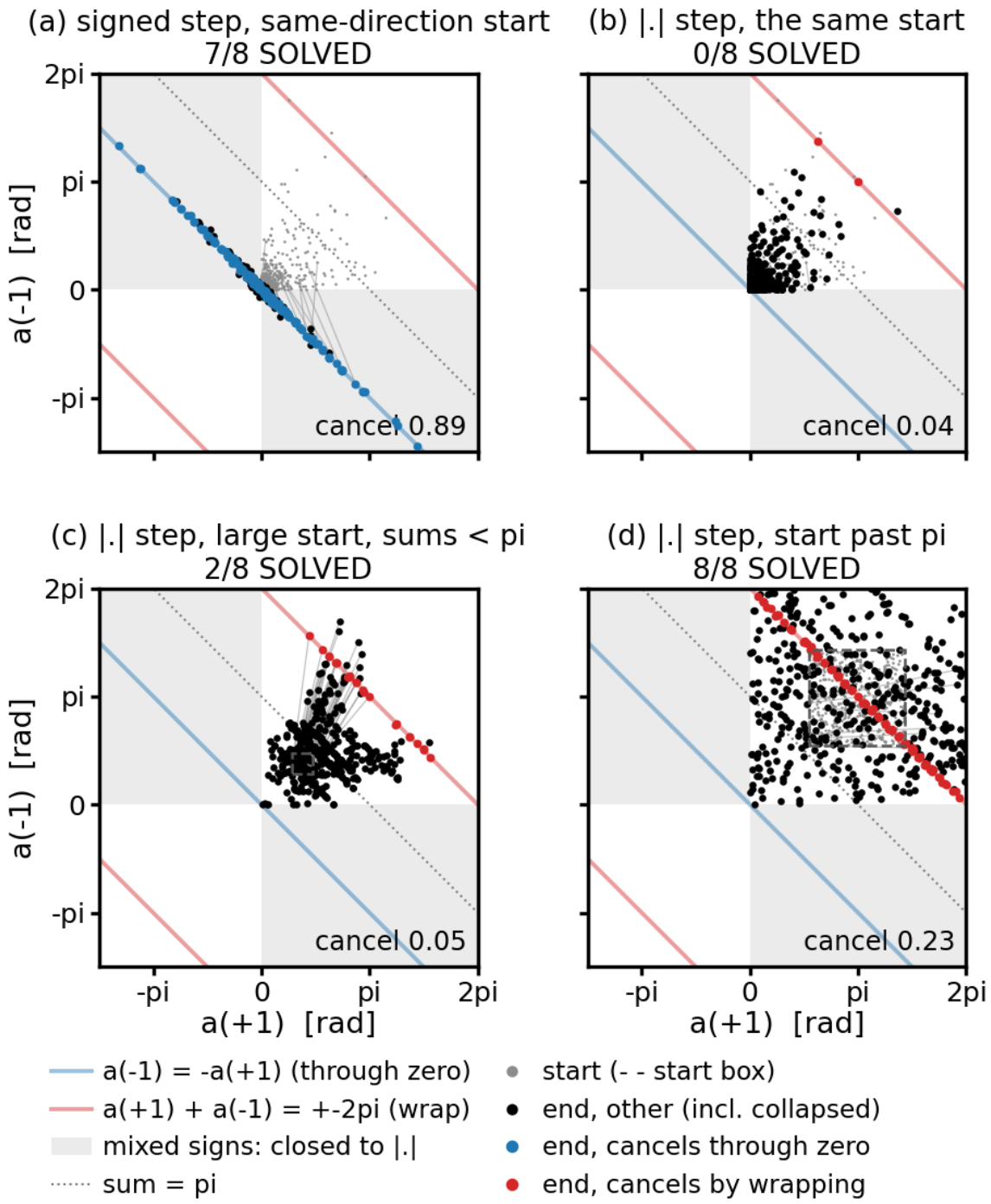


Figure 9 (`fig_mechanism`; ring, 8 runs per panel; analysis of a registered batch). Move angles (a(+1), a(-1)) of every rotary channel at the start (grey) and after training (dots). A channel cancels opposite moves when a(+1) + a(-1) = 0 mod 2 pi: on the blue line (through zero, reached only with opposite signs) or the red line (by wrapping). Grey shading marks the two mixed-sign quadrants, the only region the |.| step cannot reach (omega's sign is free, so both angles may be negative); dotted: a(+1) + a(-1) = pi; dashed boxes: start ranges. (a) Signed step from the same-sign start: 7/8 runs solve; one move's angle crosses zero, and 89% of channels cancel at the end (6% at the start). (b) The |.| step from the same start: 0/8 solve, 4% cancelling. (c) The |.| step from a large start whose sums stay below pi: 2/8 solve, 5% cancelling, all by wrapping. (d) The |.| step from

a large start whose sums exceed pi: 8/8 solve; cancelling rises from 7% to 23% of channels, all by wrapping (App. H).

7.4 Readers cannot repair a code

More computation reads a rank-2 code; it does not repair it. Looping the single block four times (identical parameters and initialisation) or using four real layers raises rank-2 accuracy from 0.894 to 0.973 and 0.990 (permutation p 0.0034 and 0.0012; SOLVED 2/8 and 5/8 against 0/8; rank 4 0.998, 8/8). By Proposition 6 neither change alters the code family; both change how a code is read. Fitting a one-pass, position-only reader to each model's own phase recovers the one-layer models' accuracy (rank 2 0.866 against 0.894; rank 4 1.000 against 0.998) but falls 0.32-0.44 short of the looped and deep models on long-gap and wrap-only revisits (analysis; App. M). Their codes are not measurably more readable by one pass than plain rank 2's (0.893 and 0.897 against 0.866, below minimum detectable effects of 0.07-0.08) and stay far from rank 4's (p 0.0002). Run with one pass, the looped weights score 0.381 (mean; 7 of 8 seeds below the 0.51 floor); the right cell is found only at passes 2-4. Only 1 of the 16 runs finds a two-dimensional code (rank 4: 8/8); half end with x and y moving the phase along one direction. Downstream computation compensates for the rank-2 deficit and leaves it in place.

Where the angle comes from and how content uses it are separate choices. Any rotary score can be written sum_c A_c cos(dphi_c + beta_c): the phase difference dphi says where, and content sets an amplitude A and, in RoPE, a phase beta. Path integration changes dphi; PoPE's polar score removes beta, and Proposition 7 then pins every kernel's peak at zero phase difference. The two predictions have opposite signs, and so do the results. On the short-walk torus PoPE's score raises the path model (MapPoPE 0.999 against MapWM 0.971) and lowers the index model (PoPE 0.679 against RoPE 0.805), an interaction of +0.154 (8/8). Probes give the mechanism (analysis; App. L). For the path model the pinned peak aligns the kernel with the target (target coherence 0.995 against 0.70 for MapWM; Fig. 10b). For the index model no content pair's kernel prefers offset 3 over offset 1 (0% of pairs, against 53-100% for RoPE, whose kernel is a comb at the retrace offsets; Fig. 10a), and 0.110 of the 0.126 accuracy gap lies on reversals whose target sits three or more steps back. Freeing PoPE's phase offset tests this directly (registered): the index model rises by +0.122 (8/8 seeds, sign-flip p 0.0078), most on multi-step reversals (+0.180), its kernel now prefers offset 3 over offset 1 in 50-100% of content pairs, with peaks at offsets 3, 4, 7 and 8, and the path model stays at ceiling (-0.018, p 0.31) while its target alignment falls from 0.987 to 0.62. PoPE's pinned offset is the lever: it helps a path phase and hurts an index phase.

A score rule repairs the readout, not the code. On the long-walk torus at rank 2 the polar score repairs short-gap revisits mostly through the readout term (target coherence 0.617 to 0.960, p 0.0002, with the code term moving +0.06) and leaves the code term low on long-gap revisits (0.80 / 0.91 under the two scores, against 0.99 at rank 4) and wrap-only revisits (0.40 / 0.48, against 0.97 at rank 4 under either score) (analysis; each stratum read on the head that carries its same-cell attention; Fig. 10c). The rank-2 phase does not place long-gap or wrapped revisits at zero phase, and no score rule can make it: under either score rank 2 solves 3/12 (registered).

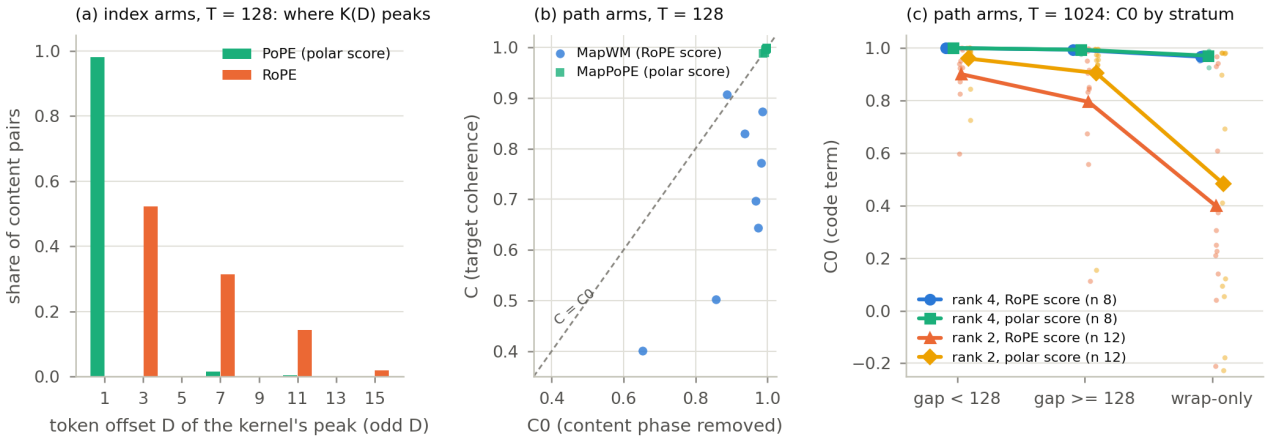


Figure 10 (`fig_kernel_peak`; analysis of registered batches). (a) Index arms on the torus at T = 128 (PAPER2X2, n = 8): the token offset at which each content pair's kernel K(D) peaks, over the odd offsets where observation keys sit (1,088 head x action x observation pairs pooled). PoPE's kernel peaks at offset 1 in 98.2% of pairs and in no run does a pair prefer offset 3 over 1; RoPE's is a comb on the retrace offsets 3 / 7 / 11 / 15 (52.3 / 31.4 / 14.3 / 1.9% of pairs; 53-100% prefer 3 over 1, per run). (b) Path arms at T = 128: target coherence C against C0, the same with the content phase removed, one dot per seed: MapWM 0.703 against 0.905 (C below C0 on 7 of 8 seeds), MapPoPE 0.995 against 0.996, on the diagonal. (c) C0 by stratum on the torus at T = 1024 (SCORE_RANK rank 2, n = 12; RANK_MI rank 4, n = 8), each stratum read on the head with most same-cell attention in that stratum: rank 2 has 0.90 / 0.96 on short gaps, 0.80 / 0.91 on long gaps and 0.40 / 0.48 on wrap-only revisits (RoPE / polar score); rank 4 0.97-1.00 in every stratum under either score. Lines: means; dots: seeds.

8. Maps from words

The text world asks whether the path model can find its actions when nobody marks them. Each move of the torus walk is written as a sentence: a direction word chosen from three synonyms, verbs that do not move the walker, optional adverbs and fillers, and, after some moves, an aside that names an object without visiting it ("she thought about a cat ."). Nothing in the input says which words move the walker (vocabulary and step tables: App. J). A walk of 1024 words is about 132 moves, so the retrace rule of Sec. 5 scores 0.934 here (analysis), beside the constant (0.505) and reversal-copy (0.597) floors.

One path layer learns navigation told in English. On fresh seeds (n = 6) one path layer scores 0.979 +/- 0.048 (6/6 SOLVED) against 0.505 for one-layer index RoPE (0/6), +0.474 (permutation and Fisher p 0.0022); two index layers reach 0.793 (0/6). With the attention-dropout evaluation scale (App. C9) the path layer scores 0.993 over all eight seeds. Both index models stay below retrace on every seed (the best two-layer seeds reach 0.927 and 0.926 against 0.934). The decisive revisits are the 13.4% that retrace cannot reach, places reached by a route other than undoing the last run (Table 7; analysis, all 200 evaluation walks): there the path layer scores 0.956 over all eight seeds (re-scored; 0.978 on the fresh seeds, 0.934 in eval mode), while two index layers score 0.411 and one index layer 0.454, both below the constant 'nothing' (0.506). Of these revisits, the 6.6% whose cell holds an object are the ones retrace gets wrong (on blank cells its default 'nothing' is right by chance); there the path layer scores 0.925 (0.967 on

the fresh seeds), two index layers 0.129, the best constant 0.088 ('nothing' excluded by construction), and one index layer 0.004, since it predicts 'nothing'. Two index layers learn part of retrace and nothing that needs the map.

Table 7. Text world: accuracy by whether the retrace rule can reach a revisit (analysis; registered batch, all 200 evaluation walks of the registered stream, 6,381 revisits; mean over 8 seeds, fresh seeds 2-7 in brackets; path models re-scored with the attention-dropout scale unless marked).

model (n = 8)	all revisits	retrace applies (n 5,525)	retrace cannot reach (n 856)	... cell holds an object (n 423)
path, 1 layer	0.993	0.999	0.956 (0.978)	0.925 (0.967)
path, 1 layer, eval mode	0.969	0.975	0.934 (0.958)	0.884 (0.927)
index RoPE, 2 layers	0.772	0.828	0.411 (seeds 0.34-0.55)	0.129
index RoPE, 1 layer	0.505	0.513	0.454	0.004
best constant	0.505 ('nothing')	--	0.506 ('nothing')	0.088 ('cat')

What the step table holds. On every seed, opposite direction words cancel once the component common to the four directions is removed (0.002-0.012), synonyms collapse to one step (cosine 1.000), and zeroing the direction steps drops accuracy to the constant floor (0.39-0.51) (declared secondaries; Fig. 11). The rest of the vocabulary is organised by how often a word occurs per move (analysis, 32 models): adverbs and fillers, whose count per move varies, get steps of 0.004-0.03 of a move; words that occur once per move (verb, seeing words, object, full stop) may carry a common tick; aside sentences get sizeable word steps that cancel within the sentence, leaving the aside's noun 0.04-0.16 of a move off the place. Words whose count per move varies get no net step; the steps that remain belong to words that occur once per move or cancel within their sentence.

The offset is why learned steps beat being told. A model whose steps are restricted to the twelve direction words stops at 0.970-0.974 on every seed, and its errors name objects told in an aside at that place: with no other word stepping, the aside's noun sits at the place's phase. Moving each aside back onto its place's phase at evaluation, with the phase after it unchanged, drops the learned-step models at ceiling from 0.989-1.000 to 0.952-0.975, with 71-100% of their errors now of the same kind; offsetting the oracle's aside words by half a move lifts it from 0.962-0.981 (on the analysis stream) to 0.999-1.000 on every seed (analysis). "Only actions move" is the wrong separation in language: some words that are not actions need steps, so that what is mentioned is not filed where it is not.

Half the models also count moves, in channels they do not use. On four of the eight registered seeds the direction words' common component points along the verbs' step (cosine +0.999 to +1.000), and 31-38 of 64 channels drift between visits to the same cell (declared secondary; drift from the first visit). Across 24 learned-step models (analysis), 10 carry such a per-move tick, which drifts in 26-38 channels between consecutive visits. It counts moves: an aside sentence adds at most 0.06 of a tick, a filler word at most

0.007. In Howard et al.'s terms these channels integrate distance travelled (Case II) beside the others' position (Case III). Proposition 2(b) says such channels cannot serve retrieval by place, and they do not: the tick alone predicts which channels drift (agreement 0.98-1.00; informative on the 10 clock models, where the predicted and observed counts of drifting channels match exactly), 79-99% of retrieval usage sits in channels whose tick is zero modulo pi (on the even torus a return always takes an even number of moves), and removing the 26-38 clock channels from the score costs at most 0.009 on each of the seven clock models at ceiling, while removing as many map channels collapses accuracy (to 0.03-0.65, means over random draws). The three clock models below ceiling still lean on their clock channels (removal costs 0.09-0.26). The clock seeds' extra dependence on non-direction words follows the direction words' common component in the used channels (Spearman +0.905 over the 8 registered seeds, exact permutation p 0.0046), which those words cancel within each move.

What must and must not move. The same rule covers the what-to-where leak on a task with new objects in every episode. A step that depends on an object's identity spreads objects at one place over identity-dependent offsets; removing the objects' steps, or normalising the step input, lets every run converge (16/16 SOLVED against 0/8 for MapWM; 8/8 against 0/8 on fresh seeds; App. K). A token's offset from its place should depend on its role, not on its identity.

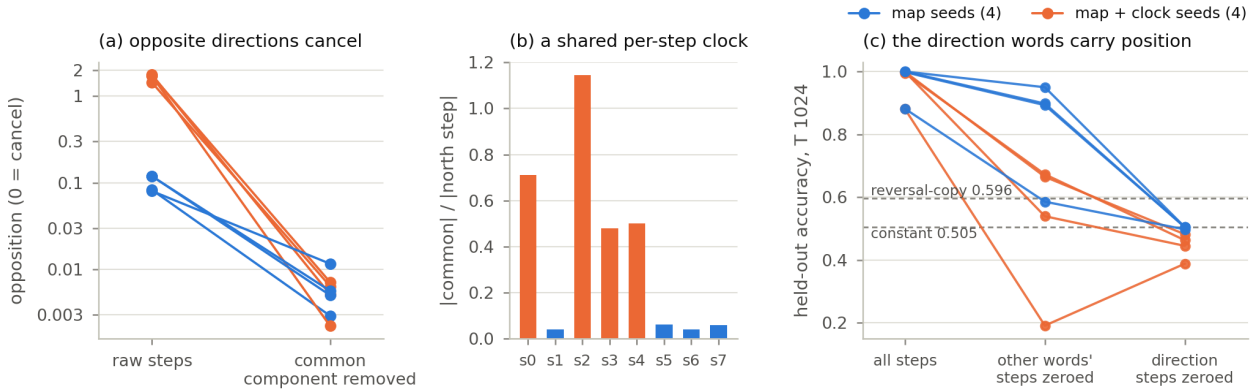


Figure 11 (`fig_textworld`). What a one-layer path model learns from a torus walk told in English (n = 8 seeds). (a) Opposition of the learned direction-word steps, |A + B| / |A| averaged over north/south and west/east (0 = opposites cancel): after removing the component common to all four directions, opposites cancel to 0.002-0.012 on every seed. (b) That common component is small on four seeds (blue, 4-6% of a direction step) and large on the other four (orange, 0.48-1.14), where it points along the verbs' step: a per-move clock beside the map (31-38 of 64 phase channels drift between visits to the same cell, counted from the first visit). (c) Held-out accuracy with every step, with the steps of all non-direction words zeroed, and with the direction words' steps zeroed: removing the direction steps drops every seed to the constant floor (0.505, dashed) or below. Eval mode without the dropout re-scale (App. C9).

9. Discussion and related work

What the two failure axes share. For per-head rank 2 and for non-negative steps, a solution in the class exists (the rank-2 projection scores 0.9955; the |.| construction is

exact), training keeps or re-finds it (7/8 at rank 2; held 8/8 on the ring and re-found 8/8 on the torus for |.|), and training from scratch does not reach it (0-2/8; 0/8). In Remark 2's terms the initialisation sits outside the basin of a non-empty, stable solution set. The route inventory accounts for this, proved for the routes and conjectured for the rank-D dynamics, and the two axes fail differently inside it. A non-negative increment has no collective route and its per-channel targets lie past pi; restoring a signed increment under a non-negative readout restores learning (16/16). At rank D the collective route exists, but no channel can be both clock-free and a full map while a clock remains, and in two or more dimensions hiding offers a free local move that, by Conjecture 1, starves the collective route; one more latent dimension adds a local move, parking, that costs the map nothing. Both failures end in the same measured place: a clock of the order of the map (or, in a few rank-2 runs, a map that has lost a dimension), and strata that keep short-gap revisits and lose the rest. Giving the rest of the network four times the computation or the parameters reads that code better and does not find the rank-2 solution: the deficit is local to the content-to-angle map. Shakerinava et al. (2026) report "representable but not learned" for non-Abelian rotations in diagonal SSMs; we find it in single-layer Abelian tasks, produced by one rank or parameterisation choice, and give its routes.

Not every search failure is of this kind. Single-origin MapEM represents k-back recency exactly (installed and frozen, 1.000) and finds the rewind per query token, and four times the budget closes most of its deficit (0.578 to 0.928; not registered), whereas rank 2 stays at 0/8 with twice the budget (App. N).

Design guidance. Keep the phase increment signed; a non-negative readout is harmless. Give each head rank above the map's dimension (D + 1). Keep observations from stepping the phase: a step that depends on what is seen moves where the model thinks it is. Measure map use where retracing no longer suffices, against measured map-free floors and by stratum: the long-gap and wrap-only revisits separate a map from a local rule, and an aggregate accuracy near 0.9 can be either. To tell whether a trained model's phase is a map or a clock, read the used-channel clock ratio, not the growth of the phase with the token index.

What and where. In the long-walk regime, the model that learns the map separates where from what, and the models that fail do not (analysis; App. K). For the signed rank-4 model at T = 1024, one shared position kernel per head times a per-pair content gain plus offset reproduces its accuracy exactly (0.998); removing token identity within type from the score costs 0.032 (0.966; 8/8 seeds lower, sign-flip p 0.008, uncorrected; Holm over the 24 contrasts 0.08); and the content-averaged score carries 92.5% of the score's within-row sum of squares (SS(P) / SS(S)). Token type (move or observation) is essential: without it the model falls to 0.113. The non-negative and index models lose 0.25-0.31 when token identity within type is removed, and a shared kernel times a gain does not restore them; within the non-negative arms, accuracy tracks the share that survives the removal (Spearman 0.84 over 16 runs, p 5e-5). Separation is a property of the solution training finds, not of the architecture. MapFormer showed the separation in the learned embeddings (their Fig. 9); these checks show that the models rely on it where a map is needed, in the sense TEM proposed.

The integrator view. A learned, signed, content-dependent rate can realise all three of Howard et al.'s cases, and which one a channel realises is set by what the readout keys

on. Split each step into the frequency-weighted mean, a clock, and a content part, and write rho_c for the clock part over the content part at the training length (Remark 3, App. E13). For targets keyed to a place reached at variable lags the used channels must be clock-free (Proposition 2(b)), and the models put them there: torus 0.064-0.079, text world 0.06-0.23 (23 of 24 learned-step models; the unconverged one 2.5; analysis). For targets keyed to a count back, the used channels are clocks: k-back recency, the same one-layer signed class, gives 21.8-22.1. One text-world model can do both at once, with distance travelled in idle channels and position in used ones. In the rotary slot the cases coincide in what they can represent (Proposition 3), so the sign result is about learnability: a non-negative step could hold a Case III code and does not find one.

Text. On ordinary text at small scale (n = 3 pilots; the ratios are analysis), the channels attention uses are clock-dominated (rho 9.9-12.3 on TinyStories, Eldan and Li 2023; 4.5-6.6 on code; against 0.06-0.23 in the text world), MapWM matches index RoPE on TinyStories (1.5053 against 1.5050 nats per token), and at matched length path integration costs on code (+0.0056 bits per byte, 3/3 seeds). The phase grows linearly with the token index there (R^2 0.994-0.999), as it does in text-world models that carry a perfect map, where the growth is carried by channels attention does not use. The used-channel ratio separates the two; the growth exponent does not.

Egocentric actions. The cumulative phase assumes allocentric moves. With turn-and-forward actions the step depends on the heading, and recoding the actions allocentrically, what head-direction cells supply, restores path integration's advantage (position effect +0.050 to +0.488, n = 8, in a short-walk recipe; not registered; App. A5).

Related work. *Optimisation, not expressivity.* Shakerinava et al. (2026) find that input-dependent rotations in unit-modulus SSMs can represent non-Abelian state tracking but do not learn it from scratch, while near-solution starts recover it. Grazzi et al. (2025) and Mamba-3 own the representability of rotations with non-negative angles; ours is the learnability cost in the rotary slot and its route account. Orientation-fixing parameterisations have a history: scoRNN fixes the +/-1 diagonal of its Cayley transform (Helfrich et al. 2018); a reparameterised weight w = |u| cannot change sign under gradient flow (Amid and Warmuth 2020, Prop. 2; Woodworth et al. 2020); the LRU initialises its phases near zero (Orvieto et al. 2023). With fixed frequencies the parameter symmetries of rotary attention are per-plane complex scalars (Wang and Wang 2025; Tran et al. 2026); a learnable step adds the reflection of Lemma 1. *Depth and recurrence.* Depth can substitute for recurrence through shortcuts (Liu et al. 2023; Yao et al. 2021), though shallow transformers learn sensitive functions unreliably (Hahn and Rofin 2024), while Selective RoPE and PaTH (Yang et al. 2025) solve parity or Flip-Flop with one layer. We give the rate at matched depth, the mechanism trained models use, and where it breaks. *Integrators.* Howard et al. (2014) for the three rate cases; GridPE for RoPE as a velocity-controlled oscillator; TEM and TEM-t for the "where" apart from the "what".

10. Limitations

- The models are small (1-4 layers, 2 heads, d_model 128-256) and the tasks synthetic; natural text appears only as a three-seed pilot. Actions are allocentric translations.

- "Not found" is relative to the stated budgets (300-1800 epochs). Learning rates were swept on the ring only. On the torus, in the 2-layer d_model 256 model, the sign gap is a quarter of the one-layer gap in accuracy and most |.| runs were still descending, so at scale part of the cost may be speed. Selective RoPE's runs were all still descending at 900 epochs. The exp and softplus increments start at a constant positive step and were run at two fixed rates.
- Rank results use two heads; our latent is shared across heads where MapFormer's is per head (per-head arms are reported). The torus exchange rate is a lower bound measured at n = 4.
- The theory of Sec. 4 was developed after the registered batches. Its proved statements concern the step parameterisation and quadratic surrogates; the hiding feedback at rank D is a conjecture, and the per-channel dynamics of |.| are not derived. The checks marked (analysis) are post hoc, on one batch per arm, without fresh-seed replication; their thresholds and statistics chosen after the data were seen are listed in App. O, and one CPU analysis (the step-map basin) is not deterministic and is reported as exploratory.
- The text world is scripted (58 words, one movement clause per move, context-free direction words), and its walks are short enough that retrace scores 0.934; the map evidence there is the revisits retrace cannot reach and the mechanism.
- No recurrent or SSM baseline.

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